Varatharuban B

Assignment – Measures of Central Tendency (MCT)

Dataset Introduction
The given placement dataset has 15 columns and 215 rows, which has the following quantitative columns.
['sl_no', 'ssc_p', 'hsc_p', 'degree_p', 'etest_p', 'mba_p', 'salary'].
Since central tendency attributes mean, median and mode applicable to numerical data we grab above columns. (Mode is applicable to qualitative data as well)

Dataset summary

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108	67.3034	66.3332	66.3702	72.1006	62.2782	288655
Median	108	67	65	66	71	62	265000
Mode	1	62	63	65	60	56.7	300000

Mean = Average
Figures for mean showing the 10^{th}, Higher Secondary and Degree marks nearly 66, indicating that the average mark is 66 and students scored a bit more for entrance test that reached around 72. Again, for MBA, again dropped to 62 as average result. Also, the average salary is around 288k.

Median = Midpoint
It will indicate the midpoint of the ordered dataset. It will be useful for any dataset having extreme values. As indicated in the above summary the mean and median values are almost the same for 10^{th}, higher secondary, degree, entrance test and MBA. If have outliers, there will be a difference between mean and median (so marks don't have suspected outliers). But there is a clear difference in salary (mean=288k and median 265k). So, we can decide that there are a lot of differences in salaries.

Mode = Frequent values
Mode will explain frequency of the value has in the dataset. The 10^{th} exam most of the students scored with 62, higher secondary exam most students scored with 63, degree exam with 65, entrance exam with 60 and MBA exam scored with 56 marks. Also, most of the students get the 300k salary as shown in the mode.

Overall, the class scored around 65 marks for all exams. So, we concluded that the class is above average.